

CS & IT ENGINEERING



Theory of Computation

DFA

Lecture No.-04



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Recap of Previous Lecture



Topic

length Condition DFA

Topic

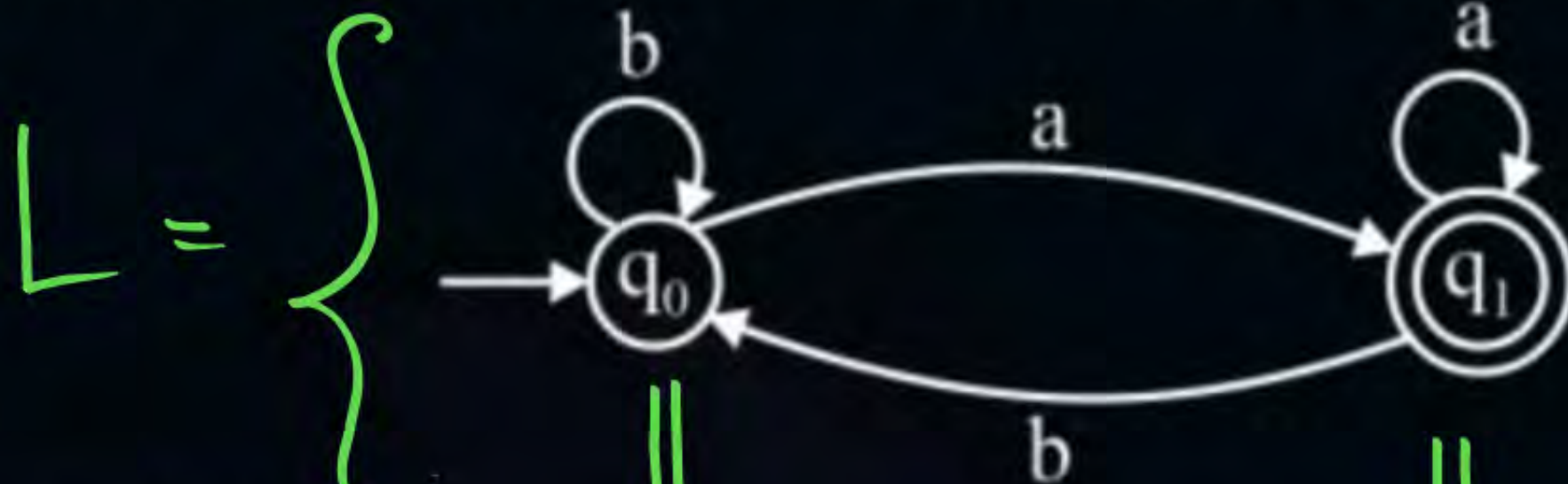
$\left\{ \begin{array}{l} \underline{(AND)} \\ \underline{(OR)} \end{array} \right\} \times \text{product DFA}$



Topic : Complement of DFA

By interchanging final and non final states we can convert into complement DFA.

{ all strings ends with a }

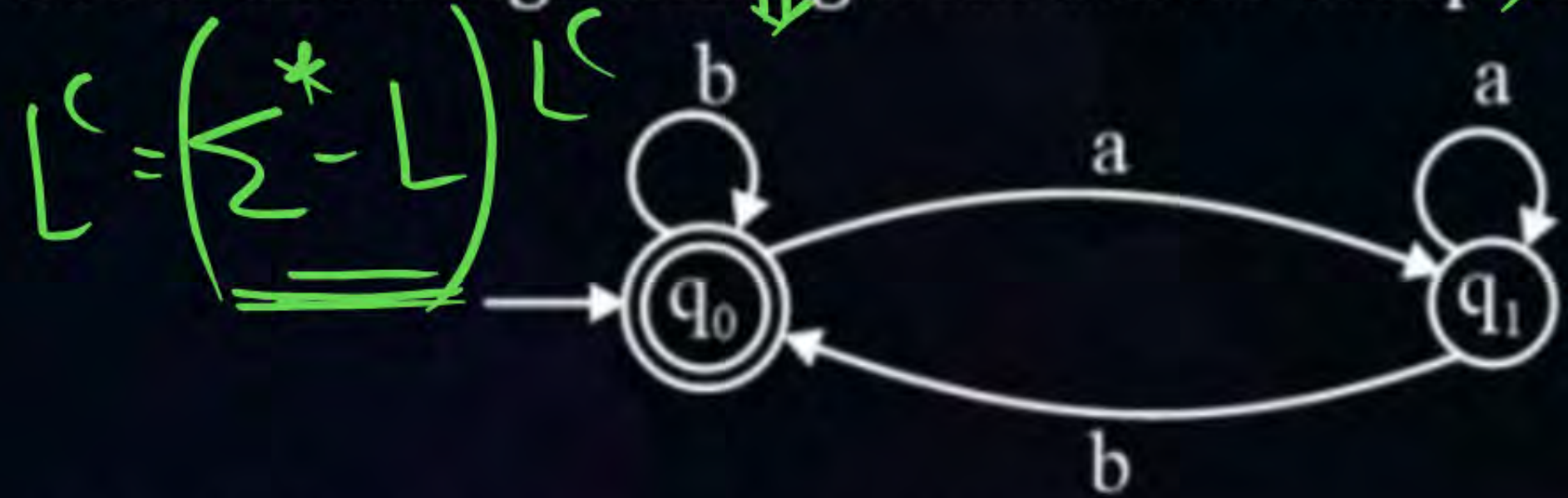


DFA?

$\{a, aa, ba, \dots\}$

not ending with a

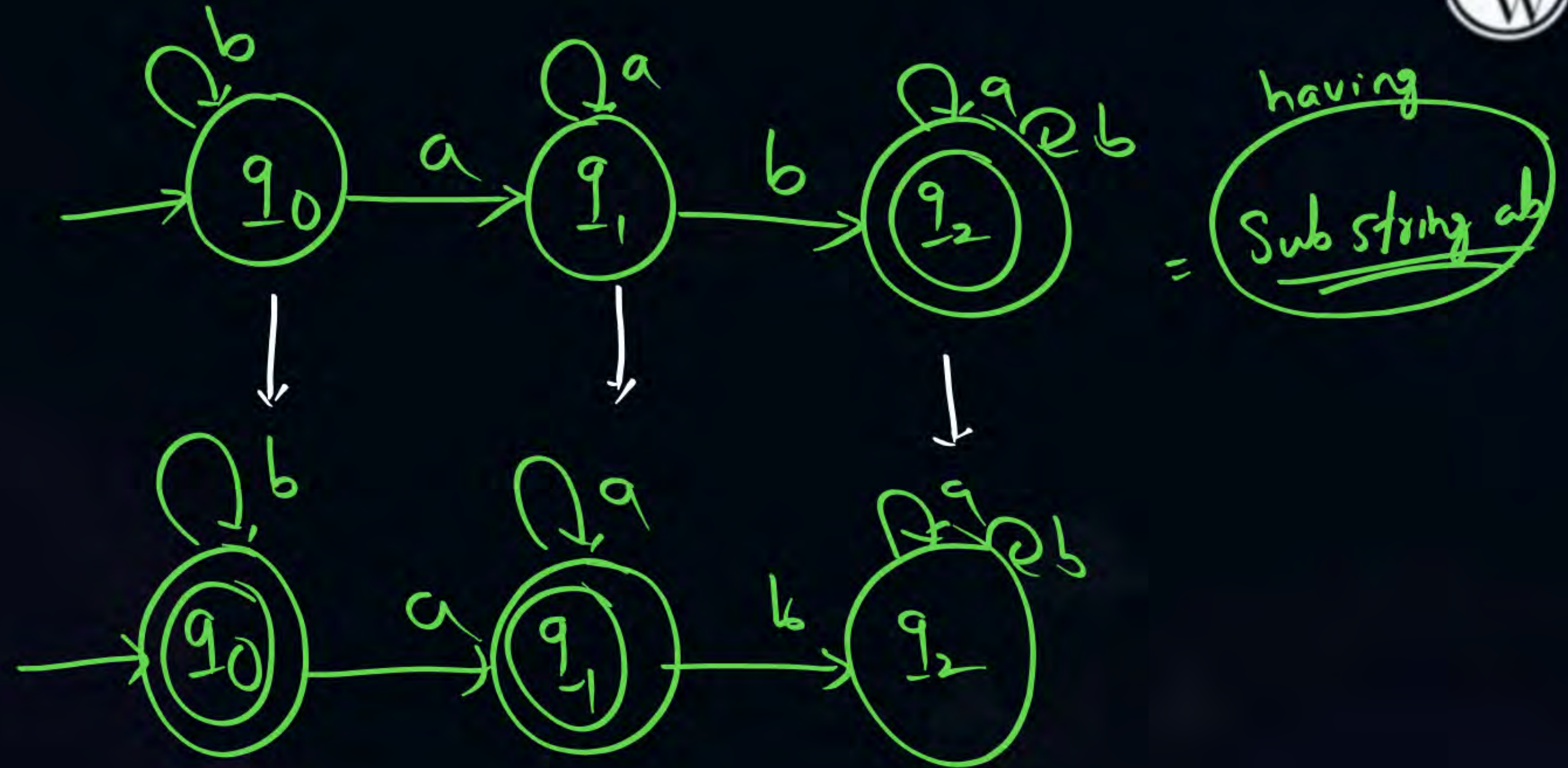
Set of all strings ending with a after complement



Set of all string Ending with

DFA?

- ~~(a) ending with b = {b, ...}~~
- ~~(b) not ending with a = {ε, b, bb, ...}~~
- (c) both a & b
- (d) none

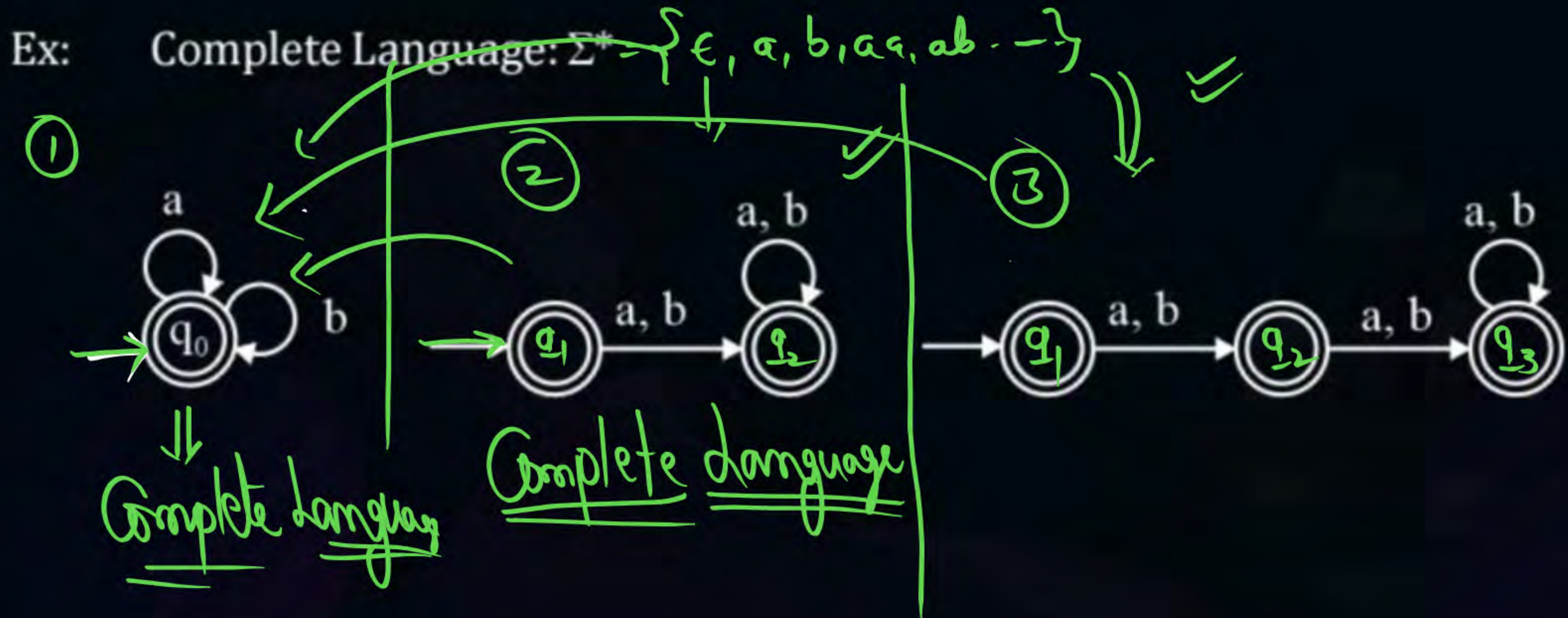


Language = { not having sub string ab }



Topic : Minimization of DFA

- For a given regular language even though many DFA exist but minimal state DFA is unique.





Topic : Minimization Algorithm

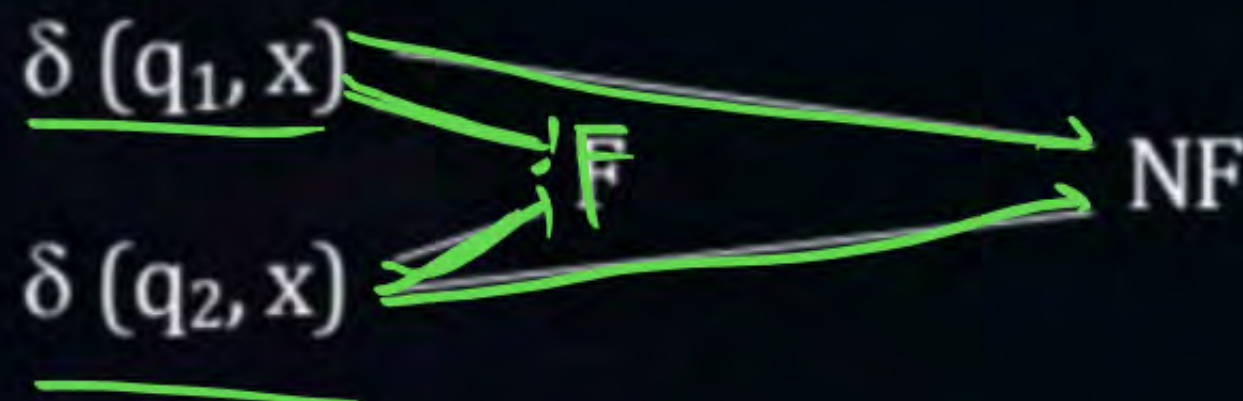
1. State equivalence algorithm

2. Table filling algorithm

Equivalent States:

Two states q_0, q_1 are said to be equivalent both $\delta(q_1, x)$ and $\delta(q_2, x), \forall x \in \Sigma^*$ should result either final state or non final state.

Distinguishable states
↓
not equivalent



$\forall x \in \Sigma^*$



Topic : Procedure of minimization

Remove all inaccessible states

- ① Elimination inaccessible states.

inaccessible state:

Any State which is not reachable from *initial* state is inaccessible state.

- ② Apply algorithm steps
3. Merge single group into one state
4. Construct new minimized DFA

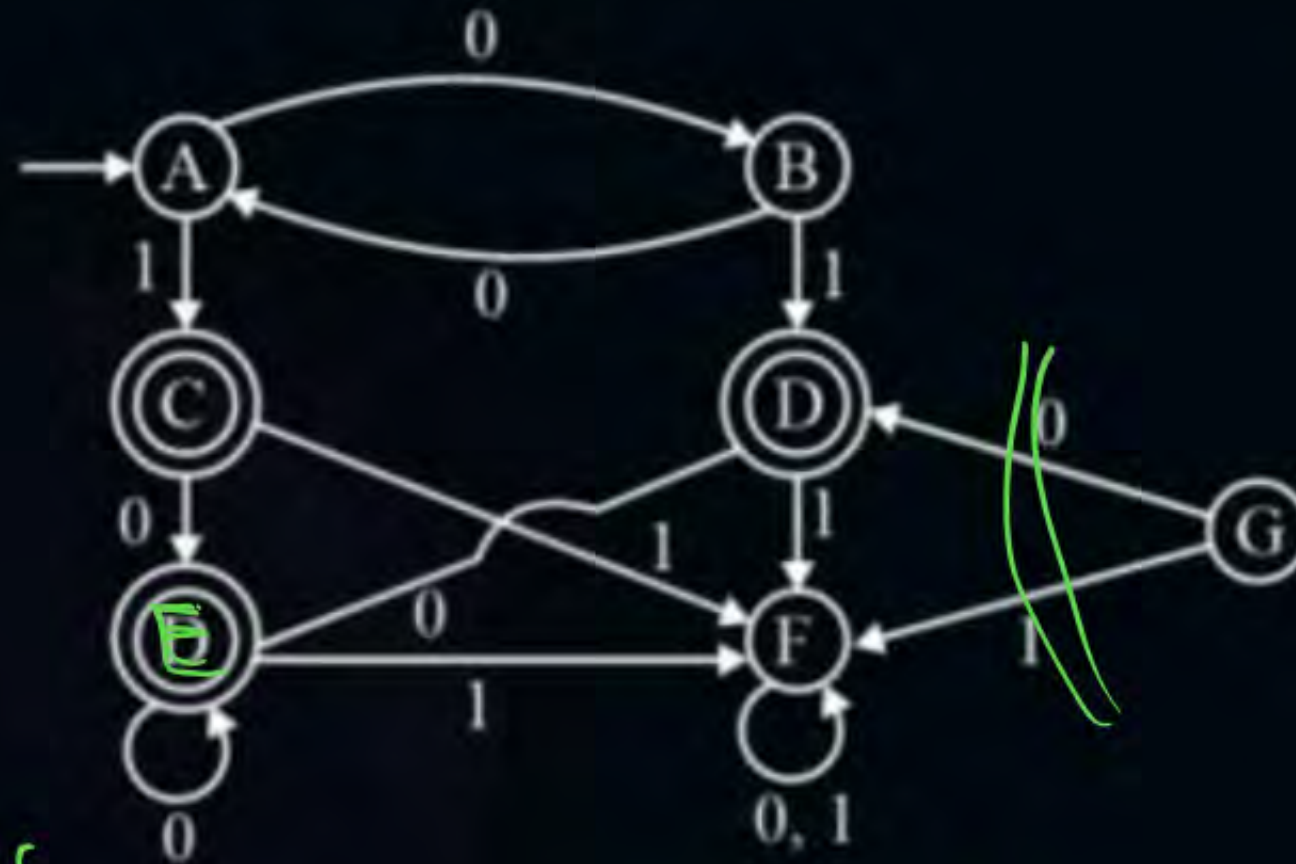


Topic : Procedure of minimization



1. Reduce states of following DFA

DFA?



Step-1: Elimination^{of} inaccessible state. = $\{G\}$

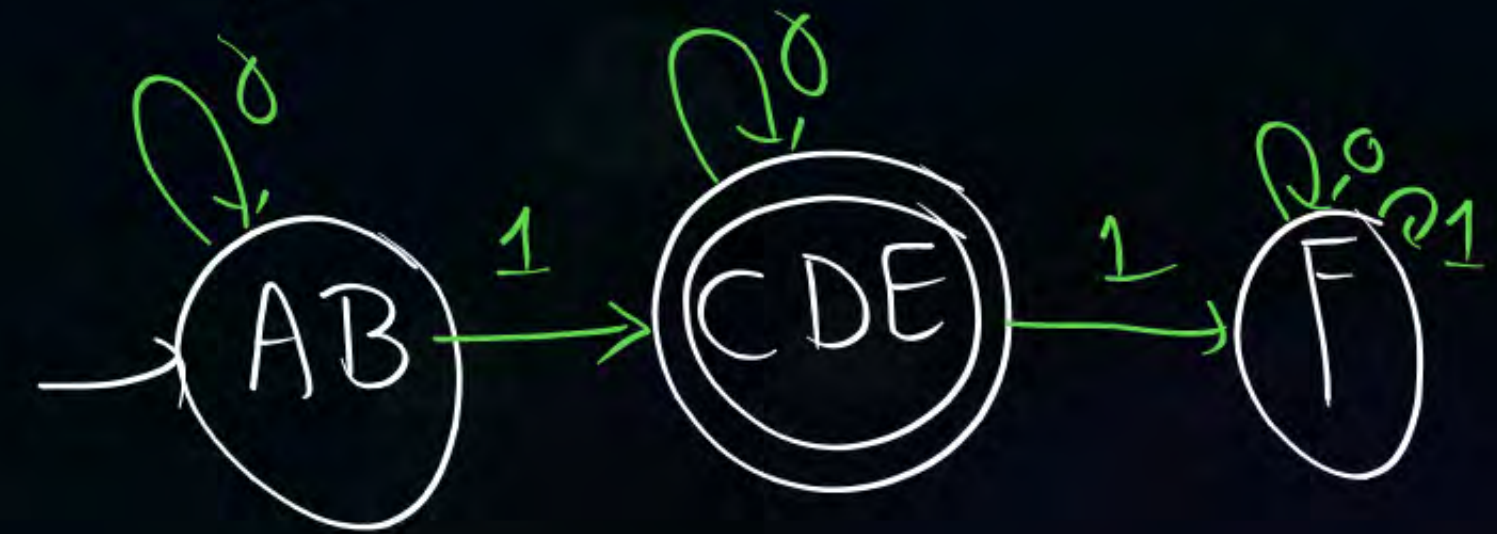
Note: Dead state is different from inaccessible state.

① Inaccessible state = $\{G\}$

② $\{A, B, F\}^{G_1} \{C, D, E\}^{G_2}$

③

min DFA





Topic : Procedure of minimization

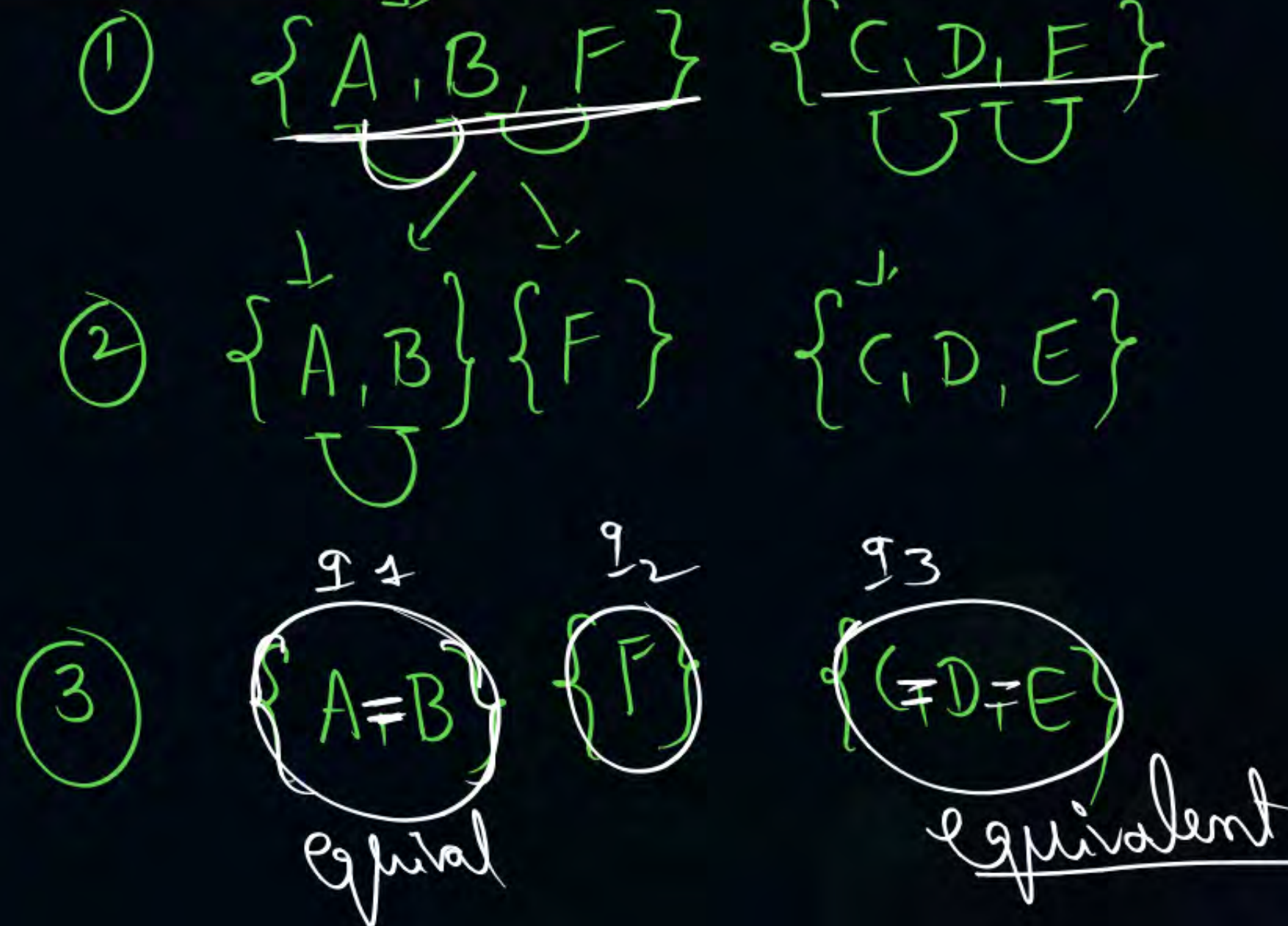
Step:2

Transition table

State	0	1
A	B	C
B	A	D
F	F	F
C	E	F
D	E	F
E	E	F

Algorithm:

1. {A, B, F} {C, D, E}
- 2.
- 3.

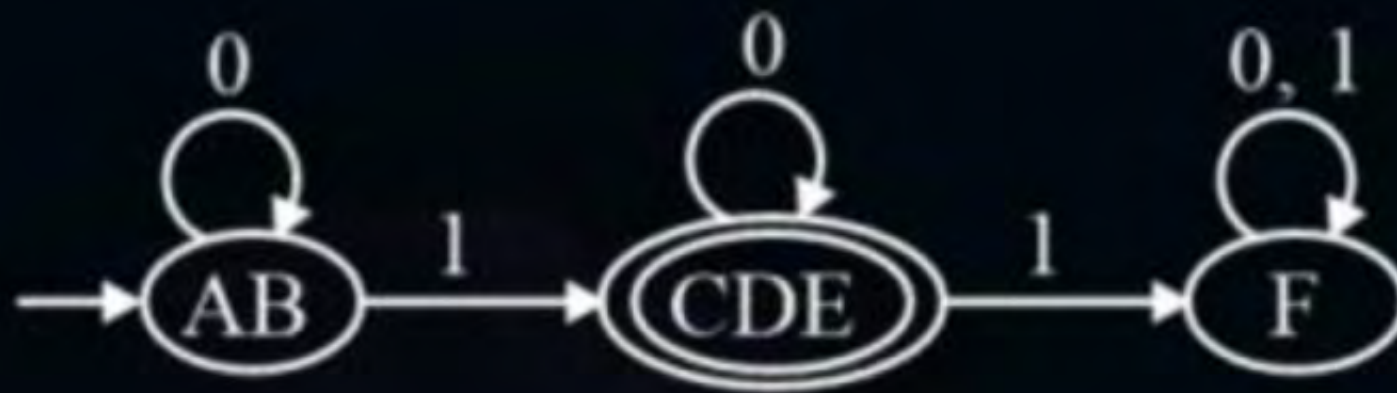




Topic : Procedure of minimization



Minimized DFA

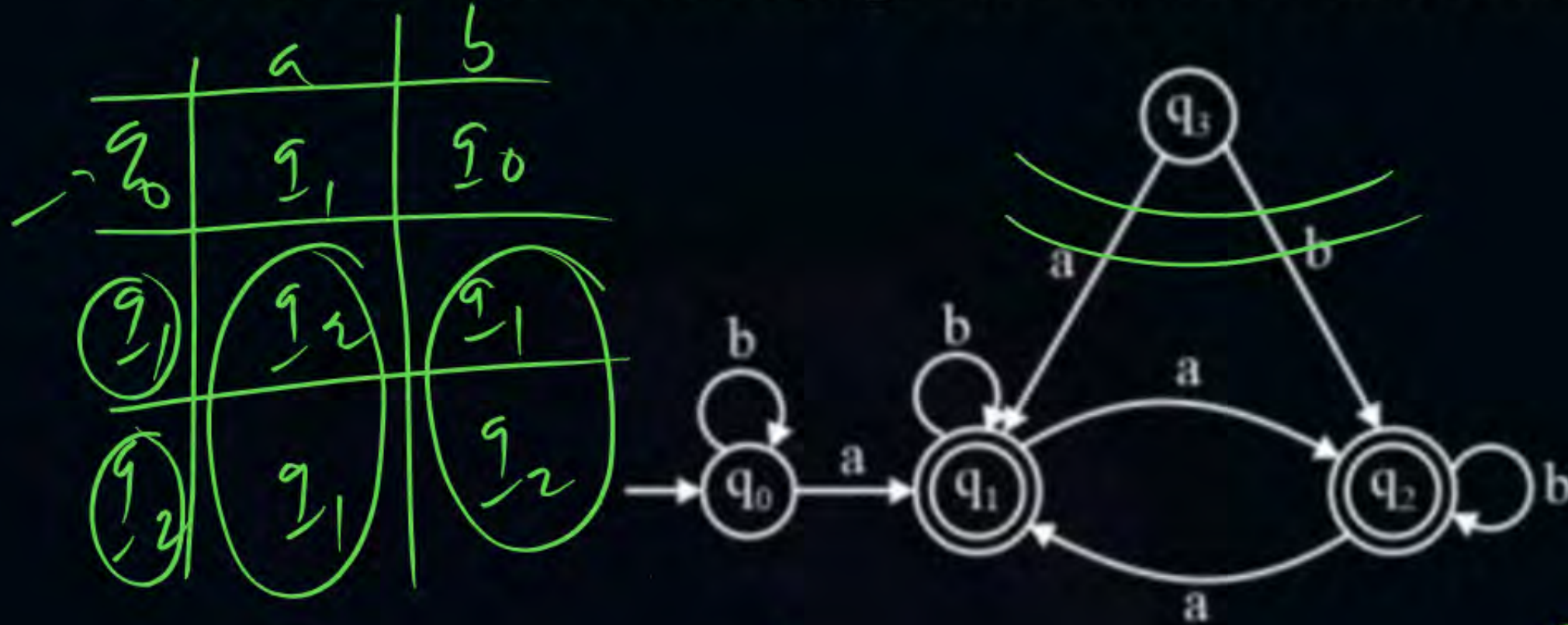




Topic : Procedure of minimization

① Inaccessible state = $\{q_3\}$

Consider the following Finite State Automation

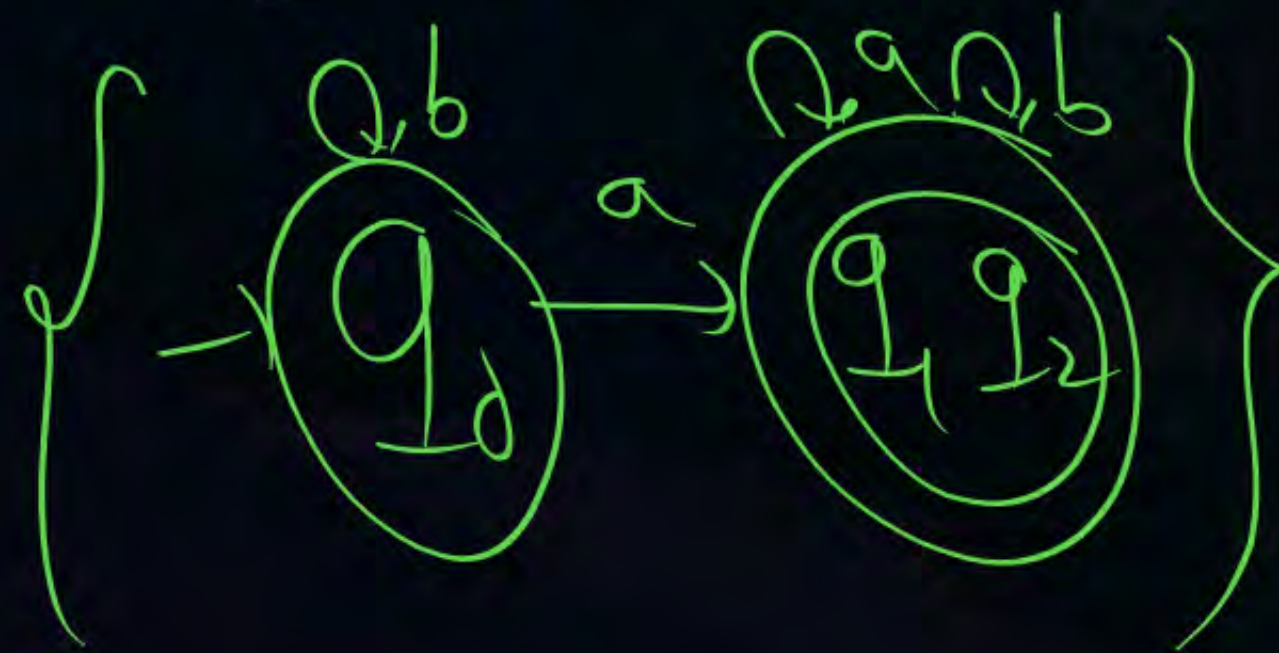


② $\{q_0\}$ $\{q_1, q_2\}$



How many states in min DFA?

2 state





Topic : Procedure of minimization



Step 1: Eliminate q_3

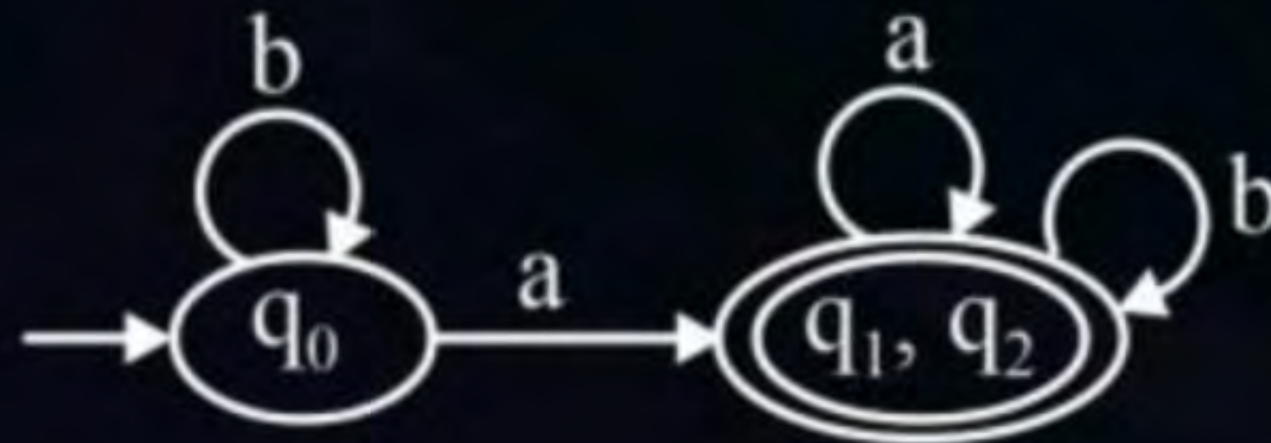
Step 2:

	a	b
q_0	q_1	q_0
q_1	q_2	q_1
q_2	q_1	q_2

Algorithm step

1. $\{q_0\} \{q_1, q_2\}$
2. $\{q_0\} \{q_1, q_2\}$

Minimum DFA

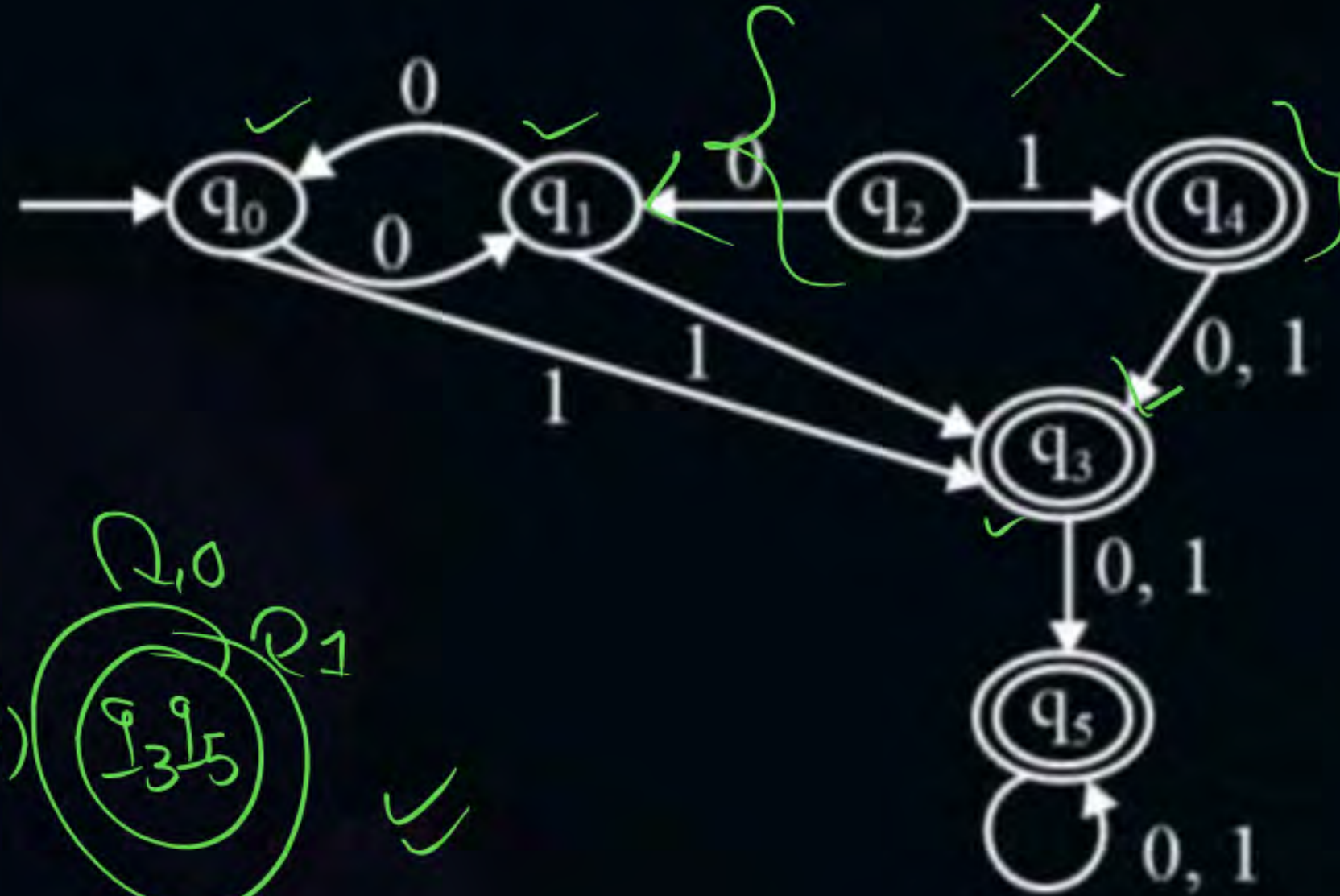




Topic : Procedure of minimization

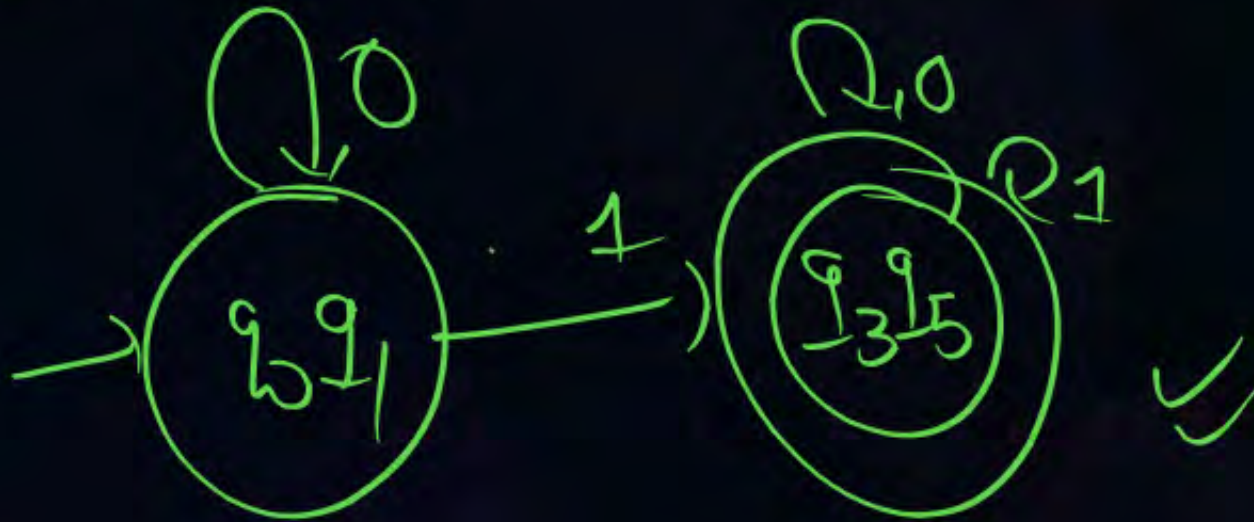
① Inaccessible state
 $\{q_2, q_4\}$

②



②

min DFA



Minimize given DFA



Topic : Procedure

Step 1: Eliminate

Step 2:

	a	b
q_0	q_1	q_3
q_1	q_0	q_3
q_3	q_5	q_5
q_5	q_5	q_5

Algorithm

1. $\{q_0, q_1\} \{q_3, q_5\}$
- 2.

Minimum DFA

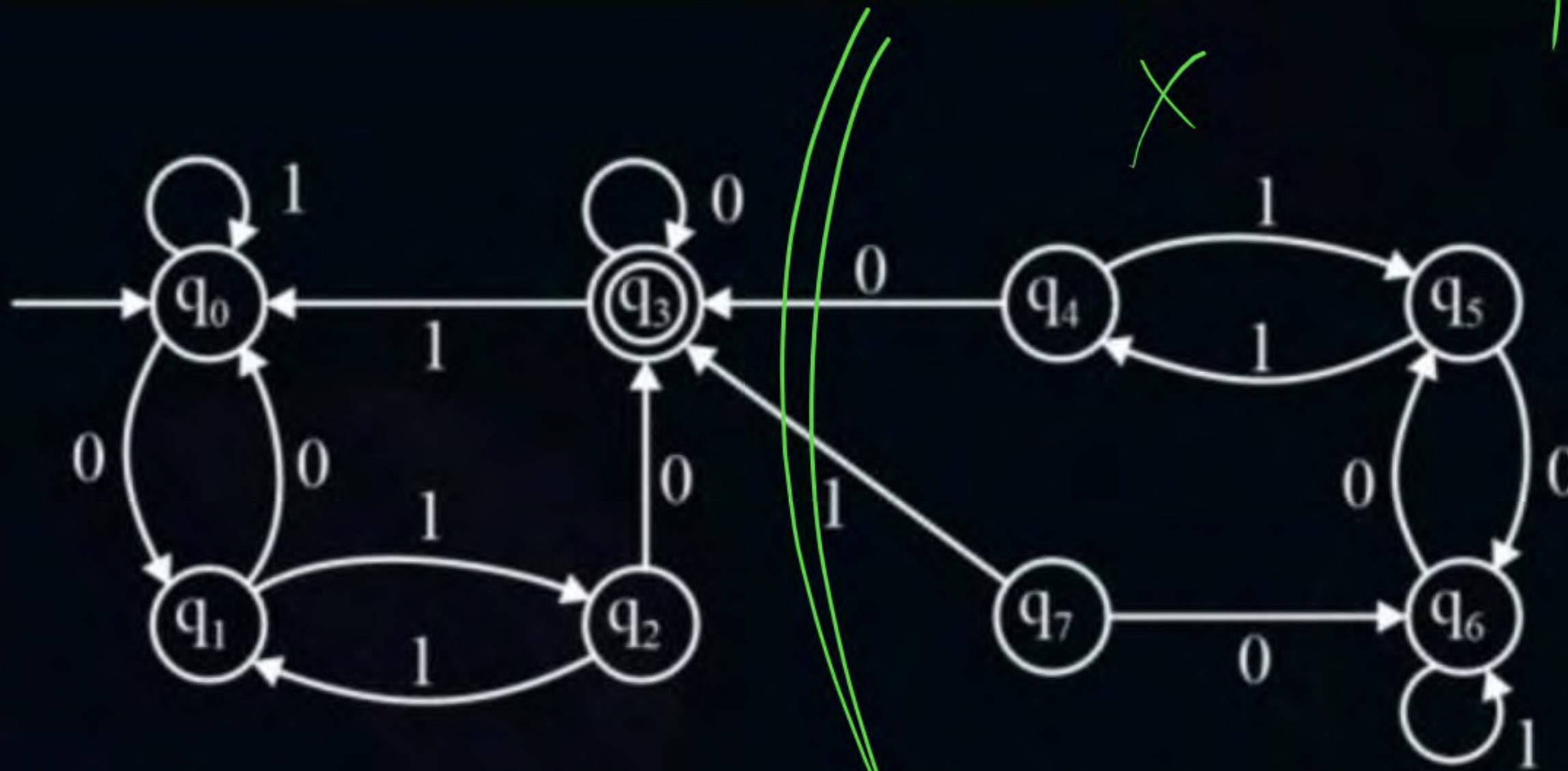
$$\textcircled{1} \quad \begin{array}{c} \downarrow G_1 \quad G_2 \downarrow \\ \{q_0, q_1\} \quad \{q_3, q_5\} \\ \cup \quad \cup \end{array}$$

$$\textcircled{2} \quad \begin{array}{c} \{q_0, q_1\} \quad \{q_3, q_5\} \end{array}$$



Topic : Procedure of minimization

$\{q_4, q_5, q_6, q_7\}$



How many inaccessible states present in given DFA



THANK - YOU